

CS & IT ENGINEERING



Digital Logic

Sequential Circuit

Lecture No. 10



By- Aditya sir



Recap of Previous Lecture



Topic



Asynchronous

↳ Full mode Counter
↳ Clear (Reset)

Topics to be Covered



Topic

Sequential Circuit

→ Asynchronous

→ clear

→ Preset





About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Ripple Counter

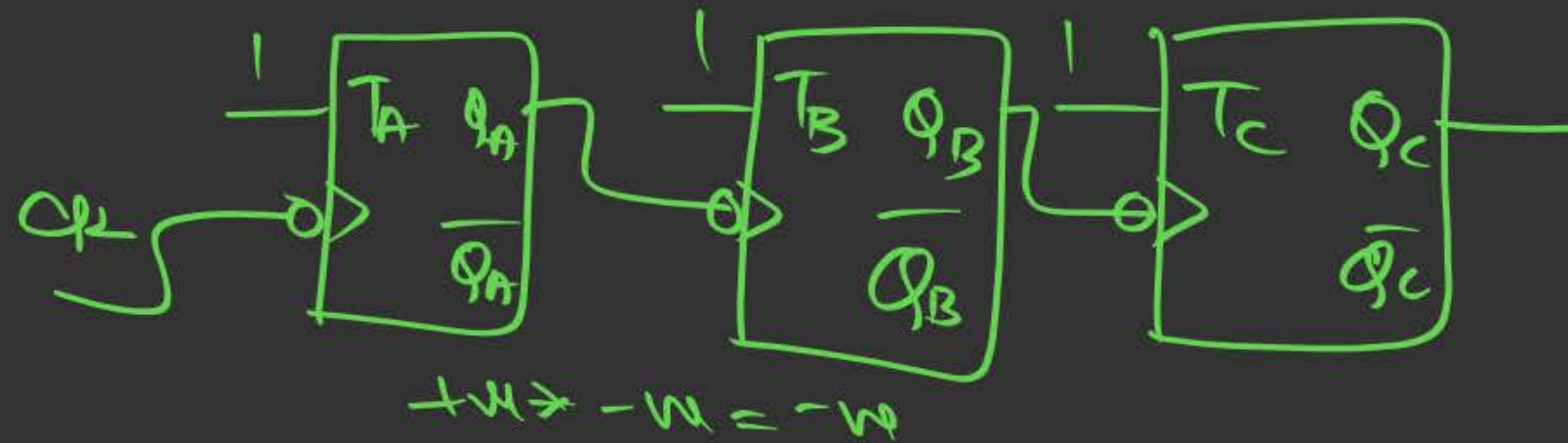
Up $\longrightarrow$ (-ve)

Down $\longrightarrow$ (+ve)

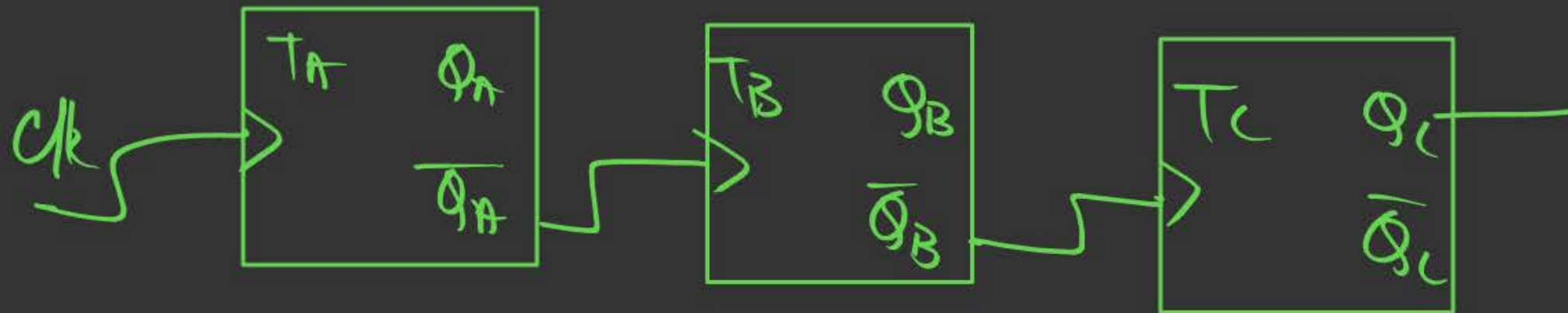
Q_0 or $\overline{Q_0}$

Up Counter

$$\underline{n=3} \quad 2^n = 2^3 = 8$$



MOD 8

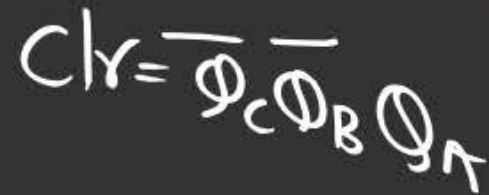


Feedback : clear (Reset)

↳ Shortcut : ① Up

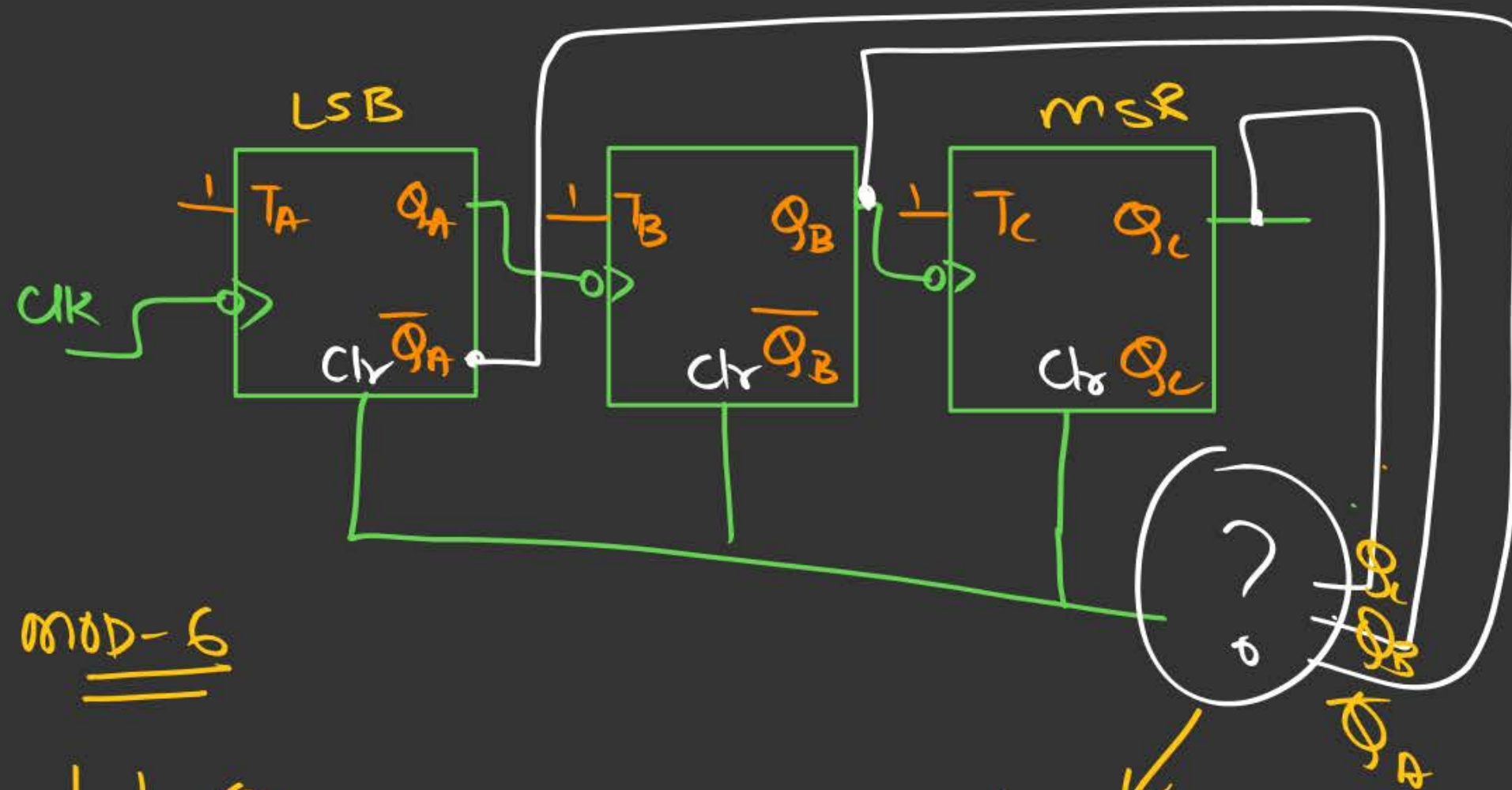
② $\text{Clear} = \text{AND}_3(Q_C \bar{Q}_B Q_A)$

(8)



$$\text{minterm} = 001$$
$$= 1$$

(Q2) Given a MOD-6 Up Ripple Counter:



Soln:-

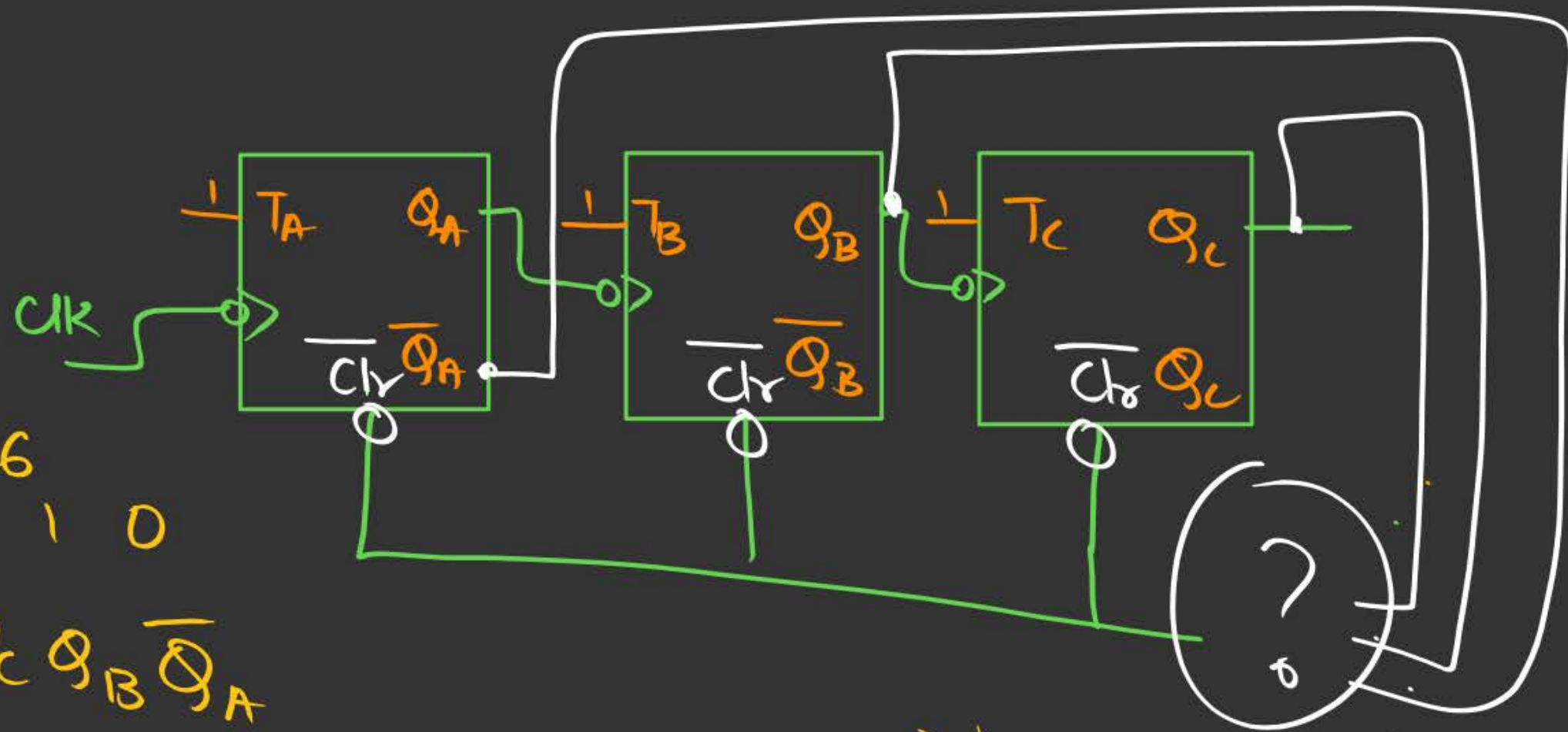
MOD-6

6 → 110

$$Clr = \boxed{Q_C \cdot Q_B \cdot \overline{Q_A}}$$

AND

(Q.3) Given a MOD-6 Up Ripple Counter:



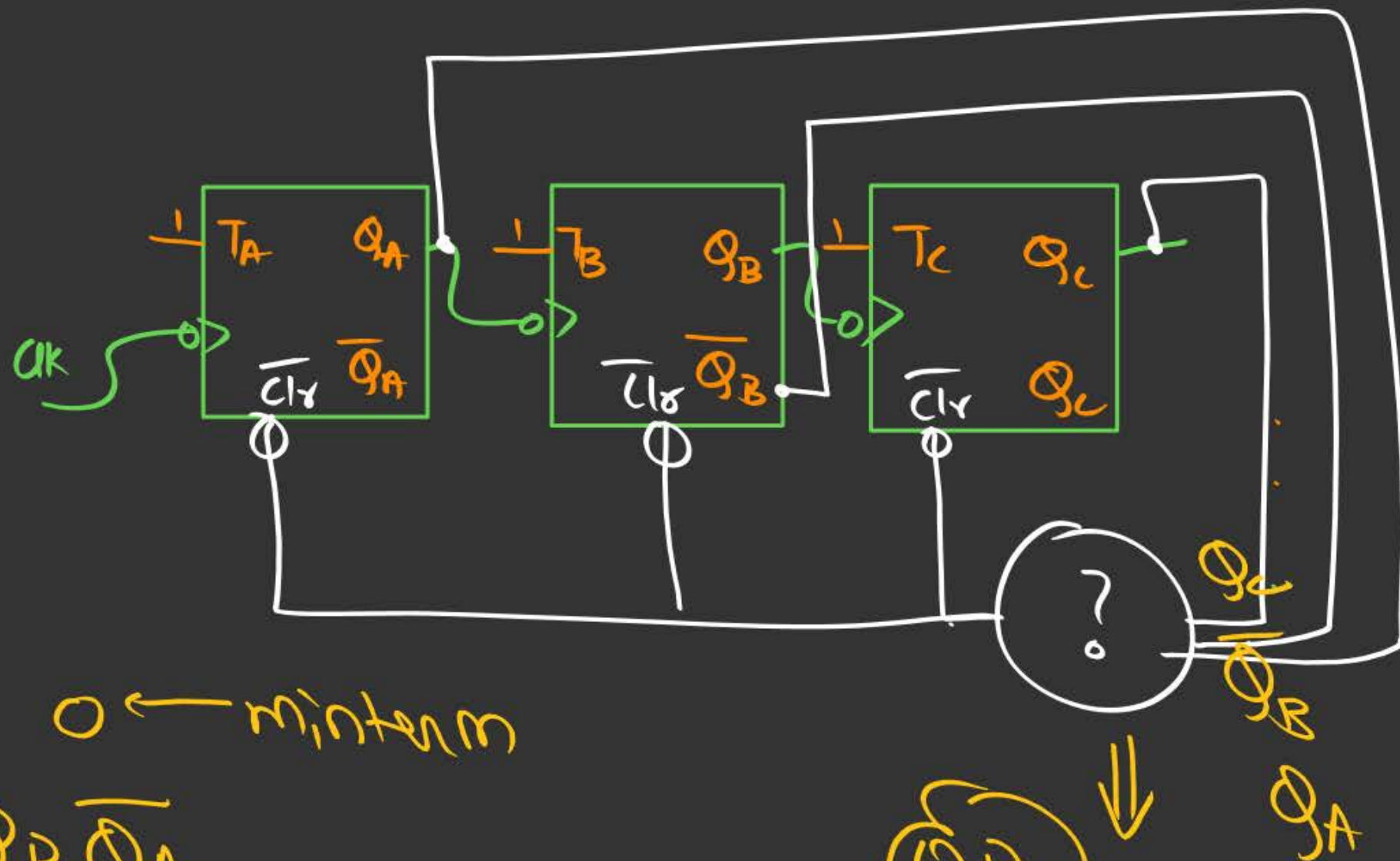
$$\overline{C_1} = \Phi_C \Phi_B \overline{\Phi_A}$$

$$C/r = \overline{Q_C Q_B Q_A}$$

(HAND)



(Q4) Given a MOD-2 Up Ripple Counter



MOD - 3

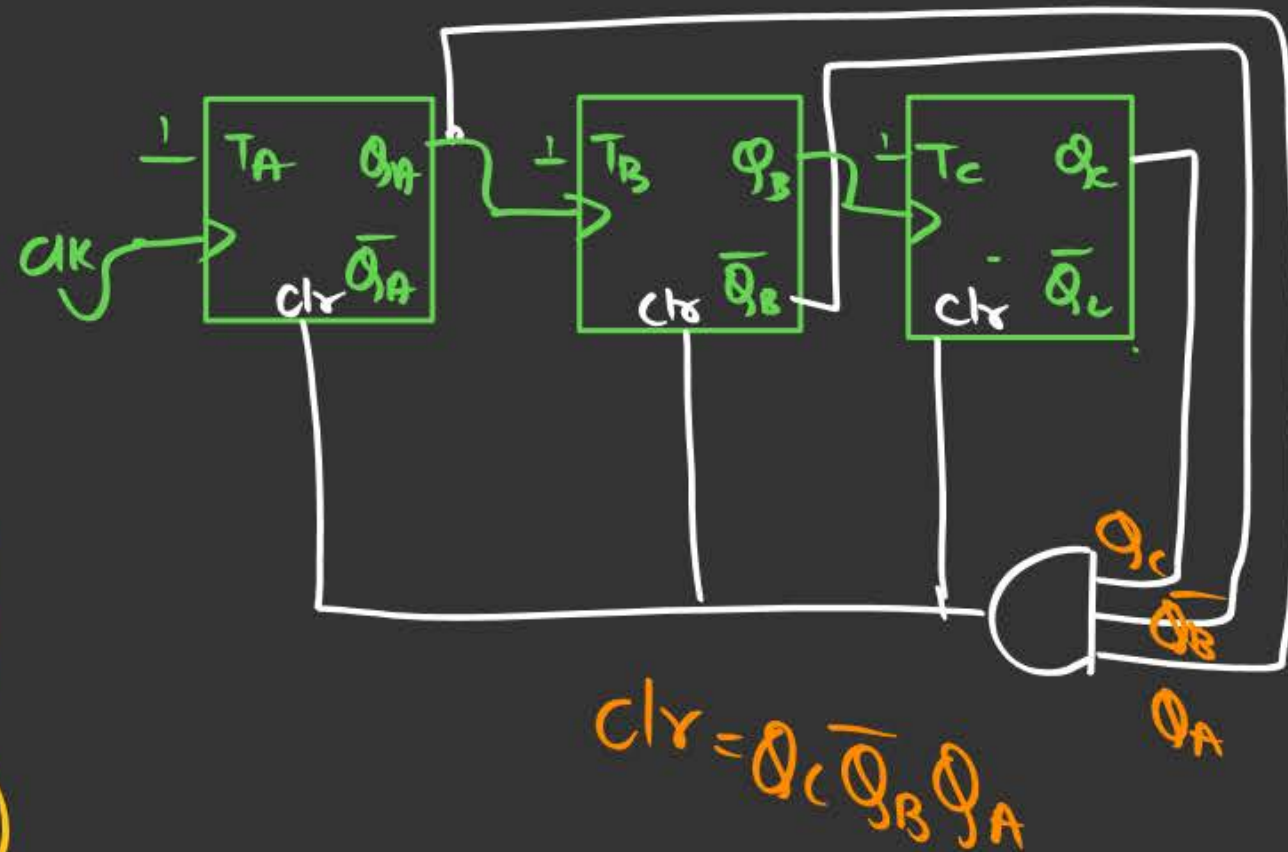
0 1 0 ← minterm

$$\overline{U}_Y = \overline{\Phi}_C \Phi_B \overline{\Phi}_A$$

$$A_1 = \overline{Q_C Q_B Q_A} = Q_C + \overline{Q_B} + Q_A$$

OR gate

MOD-3 Down Ripple Counter



$Q_C \bar{Q}_B Q_A$

clock	Q_C	Q_B	Q_A	clr
0	0	0	0	0
1	1	1	1	0
2	1	1	0	0
3	0	0	0	1
4	1	1	1	0
5	1	1	0	0
6	0	0	0	1
7				

000
111
110
mod(3)

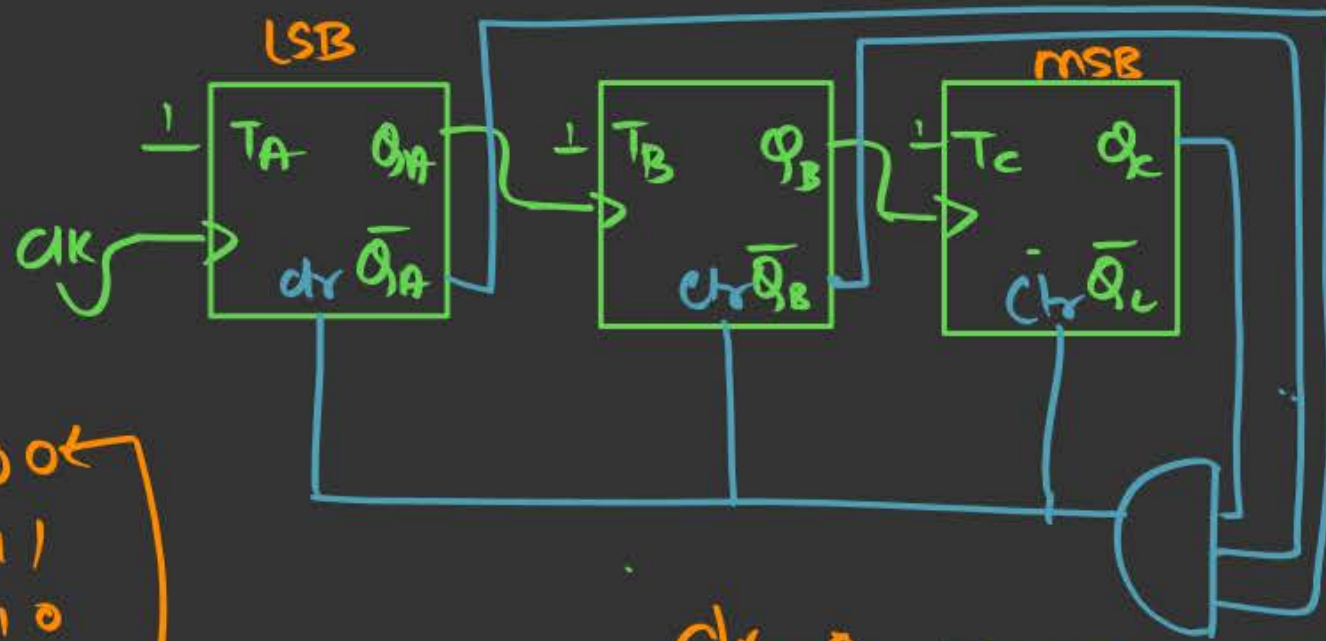
Shortcut

$$\begin{aligned} \text{clr} &= Q_C \bar{Q}_B Q_A \\ &= 101 = \textcircled{5} \\ \text{up} &\swarrow \end{aligned}$$

$$\text{Down} = \text{full-up} = 8 - 5 = \underline{\underline{3}} \quad \text{✓}$$

(Q) what type of Counter?
MOD-4 Down Ripple Counter

1
2
3
4
5
6
7



0 0 0
1 1 1
1 1 0
1 0 1

$$Clr = Q_C \cdot \overline{Q_B} \cdot \overline{Q_A}$$

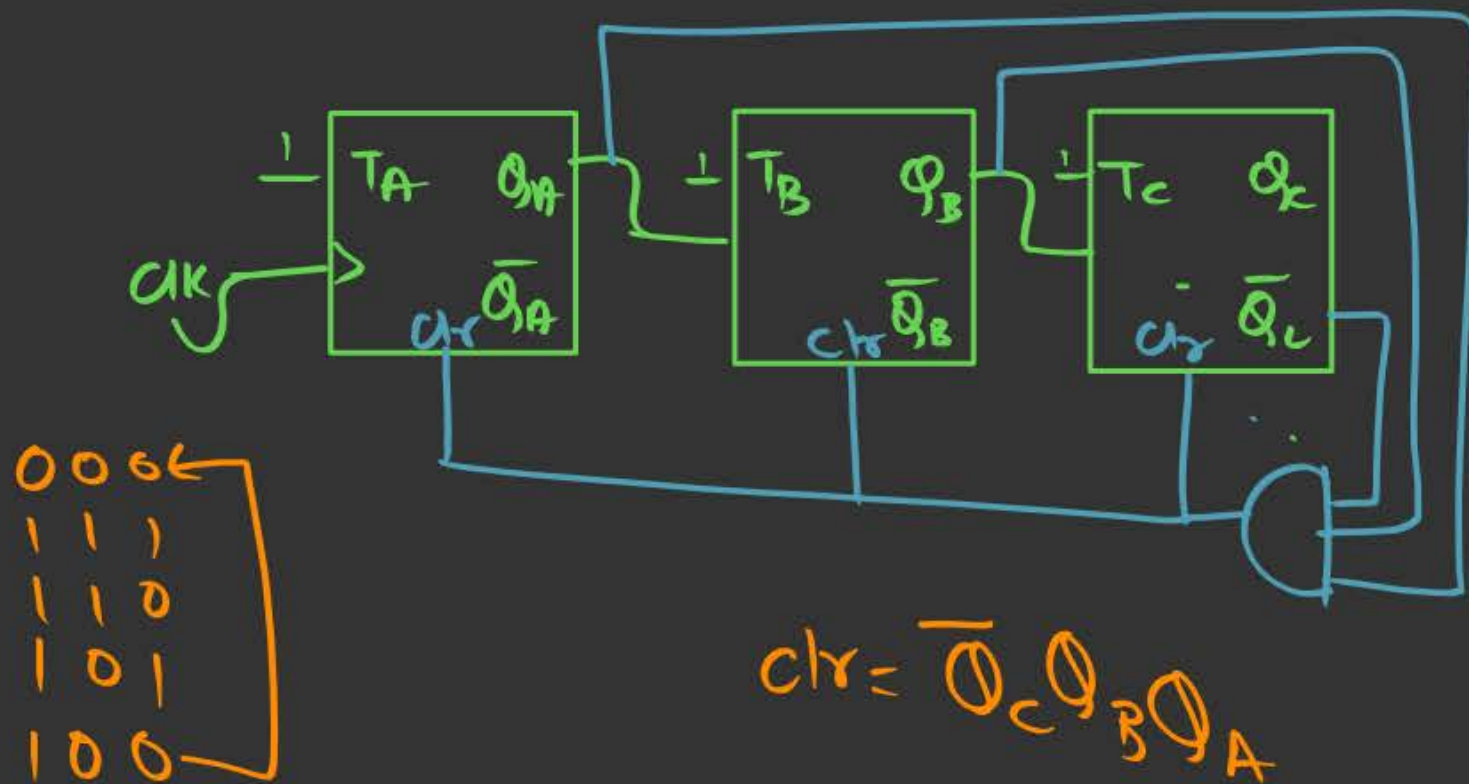
Clock	Q_C	Q_B	Q_A	Clr
0	0	0	0	0
1	1	1	1	0
2	1	1	0	0
3	1	0	1	0
4	0	0	0	1
5	1	1	1	0
6	1	1	0	0
7				

Shortcut: $Clr = Q_C \overline{Q_B} \overline{Q_A}$

minterm $\Rightarrow 100$
 $= \underline{\underline{4}}$

Down = $8 - 4$
 $= \textcircled{4}$

MOD-5 Down Ripple Counter



$\overline{Q_C} Q_B Q_A$

Clock	Q_C	Q_B	Q_A	clr
0	0	0	0	0
1	1	1	1	0
2	1	1	0	0
3	1	0	1	0
4	1	0	0	0
5	0	1	1	1
6	1	1	1	0
7	:	:	:	:

Shortcut: $clr = \overline{Q_C} Q_B Q_A$

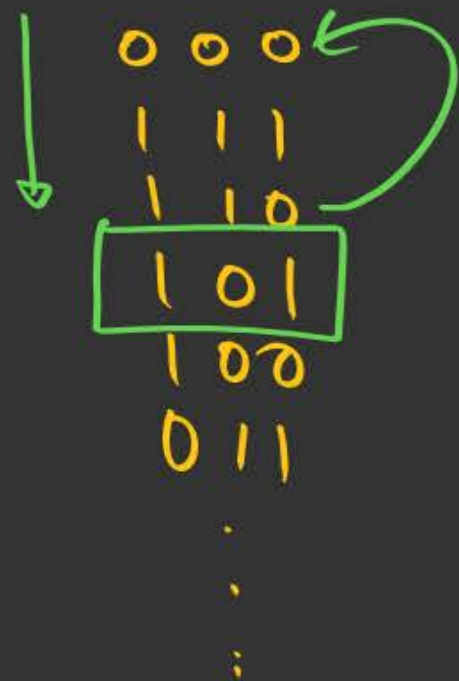
min term = 011 = 3

Down = $8 - 3 = 5$

1 7
2 6
3 5
4 4
5 3

Shortcut 😊

For Down Ripple Counter
(AND)



$$Clr = Q_C \bar{Q}_B Q_A$$

min term = 101 = 5

If UP Counter then $\Rightarrow$ MOD-5

If Down Counter then $\Rightarrow$ MOD-3

$$Full = 8 \quad (2^3 = 8)$$

$$Down = 8 - 5 = 3$$

010
😊

DOWN = FULL - UP

Challenge-1:

Design a MOD-13 Down Ripple Counter

Soln:-

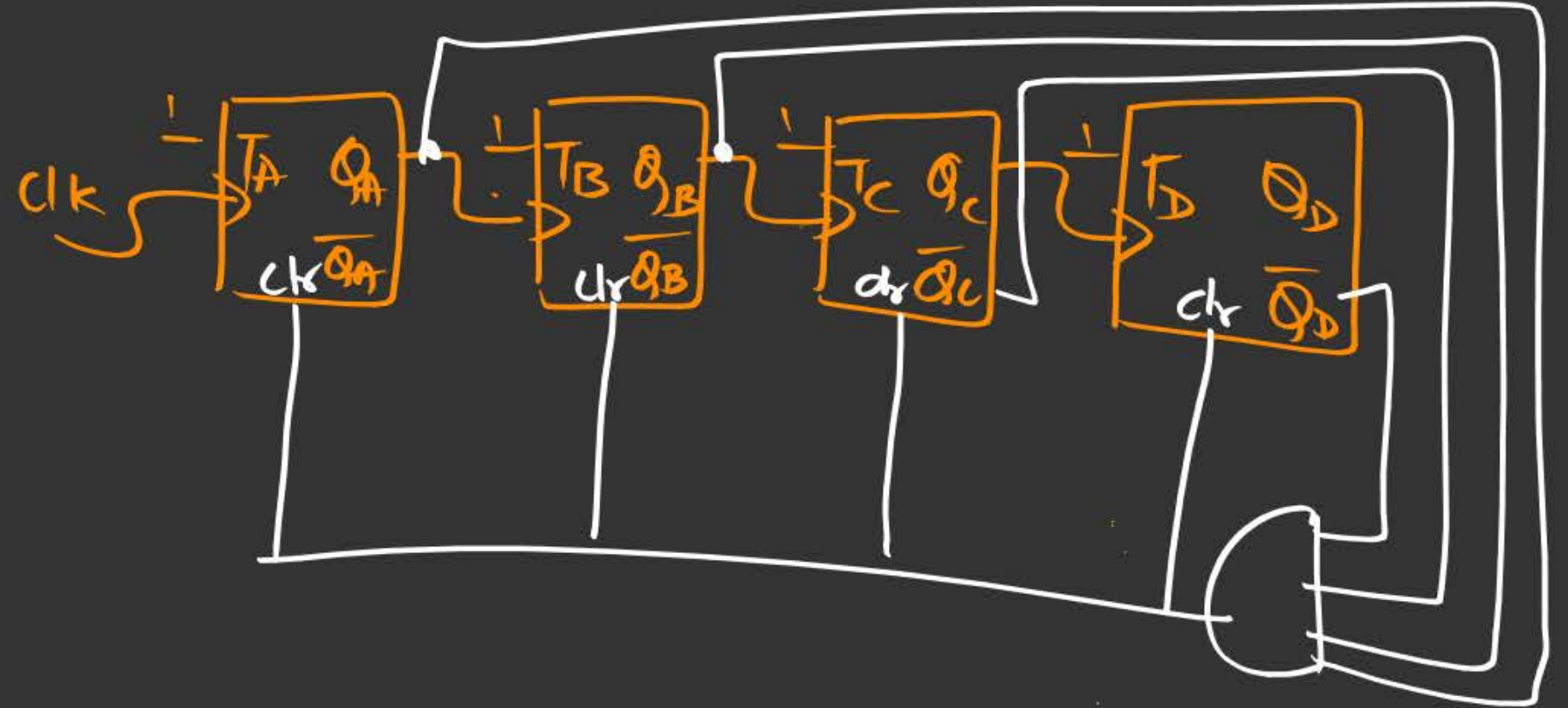
MOD-13 Down

Full = 16 (4 FFs)

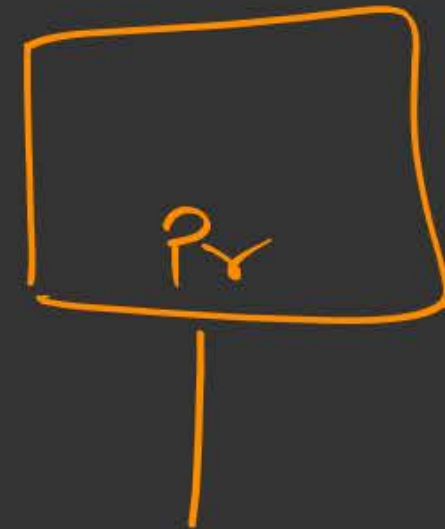
UP = Full - Down
= 16 - 13 ☆

UP = 3 → 0011

Cr = $\overline{Q_D} \overline{Q_C} Q_B Q_A$ ☆

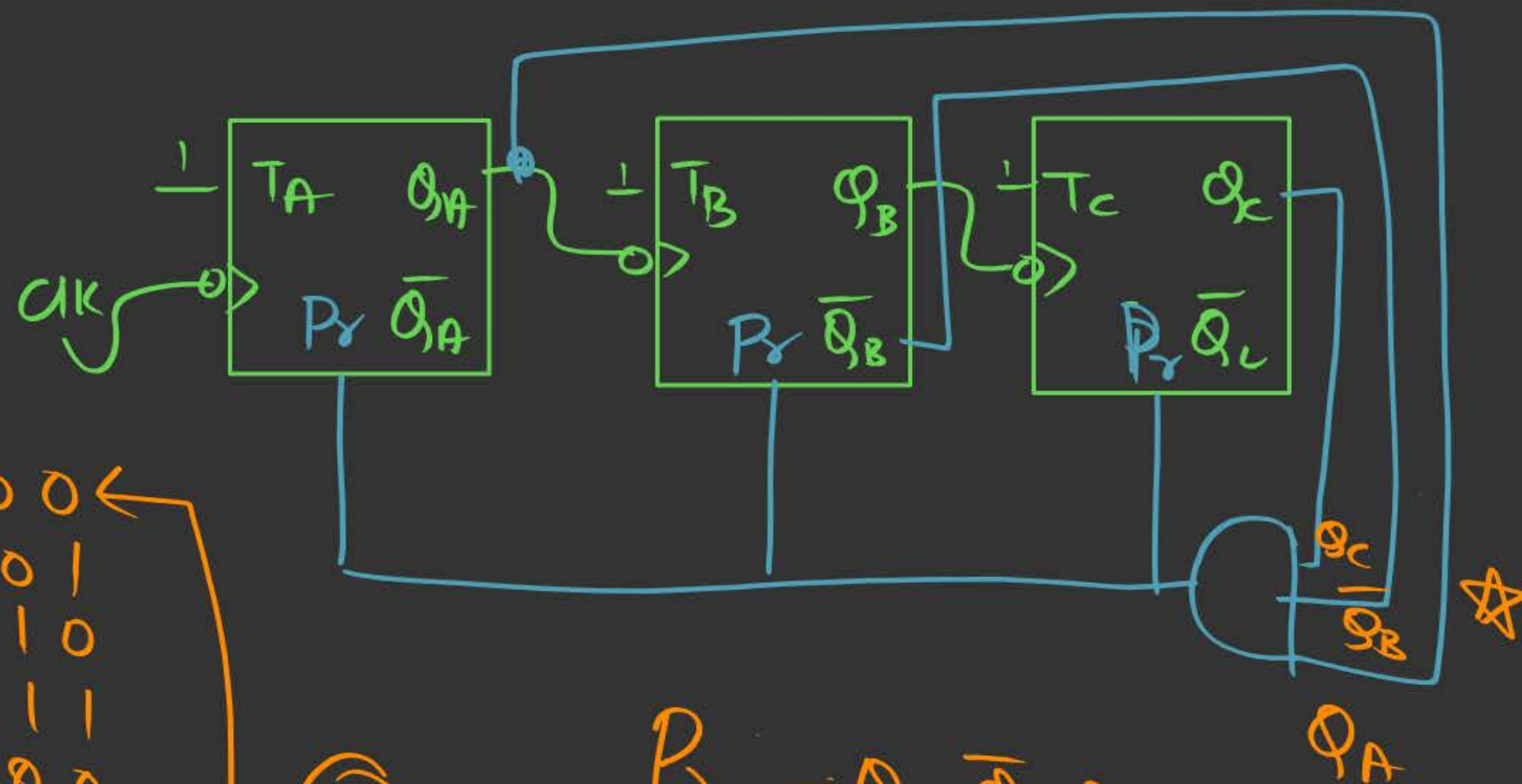


Concept of Preset



within same clock $\Rightarrow$ turn every $\phi=1$

MOD-6 Up Ripple Counter



$$P_r = Q_c \overline{Q}_B Q_A$$

Clock	Q_C	Q_B	Q_A	P_z
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1 0
6	0	0	0	0
7				

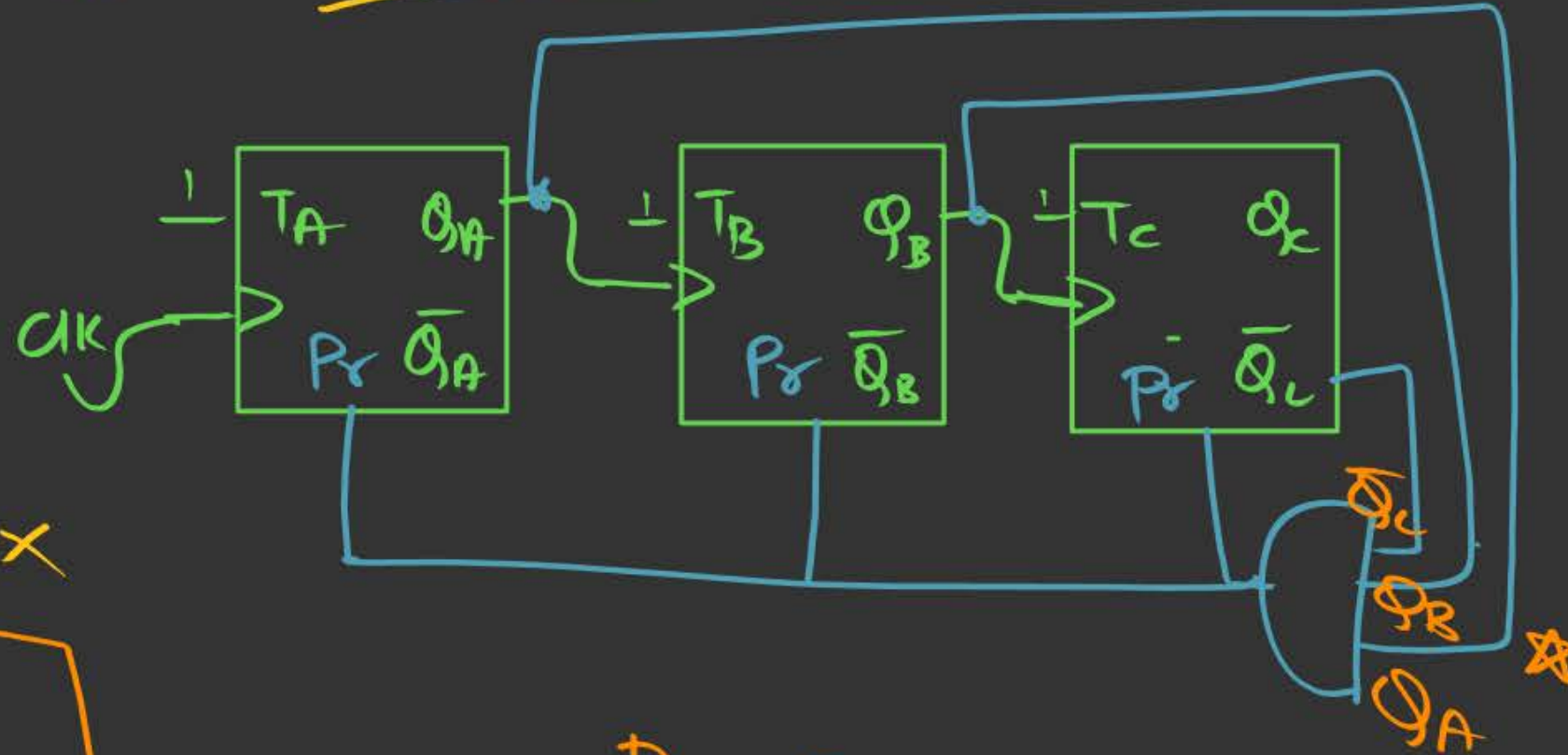
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	1	1

6

Challenge : most trick Question till now :
 ★ MOD-4 Down Ripple Counter

0/0
 00

$\overline{Q_C} Q_B Q_A$



000x
 ✓ 111
 ✓ 110
 ✓ 101
 ✓ 100

$$Pr = \overline{Q_C} Q_B Q_A$$

Clock	Q_C	Q_B	Q_A	Pr
0	0	0	0	0
1	1	1	1	0
2	1	1	0	0
3	1	0	1	0
4	1	0	0	0
5	0	1	1	1
6	1	1	0	0
7				



2 mins Summary



→ Asynchronous
Completed





THANK - YOU